



Hasan Bakır

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SUMMARY

AI & Machine Learning Engineer with more than 1 year of hands-on experience developing and deploying production AI systems in industrial environments. Experienced in computer vision, deep learning, LLM/RAG pipelines, anomaly detection, and model deployment. Built end-to-end AI solutions covering data processing, model development, real-time inference, API integration, optimization, and production deployment using Python, PyTorch, OpenCV, FastAPI, ONNX, TensorRT, and Docker. IEEE-published researcher and NASA Space Apps Global Nominee

EXPERIENCE

AI Software Developer

Feb. 2025 – Feb. 2026

AYGAZ – Koç Group

Istanbul, Turkey

- Engineered a **multi-camera tower monitoring system** for real-time truck and LPG cylinder counting across the factory yard, streaming results through a REST API into a **production-grade dashboard**.
- Developed a **vehicle type and license plate recognition pipeline** for factory-entry logging, achieving **~100% accuracy** with custom-trained YOLO detection and OCR post-processing, visualized in a real-time interface.
- Built a **24/7 production-line monitoring system**: automatically counted and logged all manufactured LPG cylinders, delivering analytics dashboards to production personnel.
- Designed and deployed an **Agentic RAG pipeline** using LLaMA and Mistral (via Hugging Face) with a natural-language-to-SQL engine, enabling intelligent database querying through a chatbot interface.
- Applied **Isolation Forest** anomaly detection for manufacturing defect identification; developed Pseudo Labeling workflows to expand training datasets with minimal manual annotation cost.

AI Software Developer Intern

Dec. 2024 – Jan. 2025

AYGAZ – Koç Group

Istanbul, Turkey

- Optimized real-time detection and tracking models (**YOLOv11**, **ByteTrack**) with **ONNX** and **TensorRT** for GPU-accelerated inference, achieving **30–45 FPS** on production camera systems.
- Built an **automated packaging inspection system** for Pürsu product lines: integrated OpenCV and YOLO with dry-contact relay hardware to trigger production-line halts on missing-item detection.

Computer Vision Engineering Intern

Oct. 2024 – Nov. 2024

AISOFT

Sakarya, Turkey

- Developed image stitching algorithms for a TÜBİTAK-supported project: implemented frame alignment, feature point detection (ORB/SIFT), and perspective correction using homography estimation.
- Built data analysis and feature extraction pipelines from structured Excel datasets; delivered insights through automated graphical visualizations using Pandas and Matplotlib.

EDUCATION

Sakarya University

Sakarya, Turkey

Bachelor of Science in Computer Engineering – GPA: 3.02/4.00

2022 – 2026

TECHNICAL SKILLS

Programming: Python, C++, C, SQL, Java

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Isolation Forest, BERT

Computer Vision: OpenCV, YOLO, Faster R-CNN, RF-DETR, U-Net, SAM, ByteTrack, YOLO-Pose, OCR (EasyOCR, Tesseract), Pseudo Labeling

Generative AI & LLMs: LLaMA, Mistral, RAG, LangChain, Hugging Face, Natural Language to SQL

Deployment & MLOps: FastAPI, Docker, ONNX, TensorRT, Git, Azure DevOps, Linux, SeqLog

Data & Tools: NumPy, Pandas, Matplotlib, Ultralytics, Roboflow, LabelImg, Anaconda

Embedded & IoT: Raspberry PI 4, ESP32, BLE, C++, IMU, Flutter, Firebase

PROJECTS

- Basketball Game Tracker Pipeline** | *RF-DETR, ByteTrack, SAM, PARSeq, YOLO-Pose, TensorRT* | [GitHub](#) 2026
- Built a multi-modal sports analytics pipeline: player detection (RF-DETR), persistent tracking (ByteTrack), jersey number OCR (PARSeq), pose estimation (YOLO-Pose), and instance segmentation (SAM + Cutie).
 - Computed bird's-eye-view homography via RANSAC on court keypoints for spatial player positioning analytics.
 - Accelerated inference with **ONNX and TensorRT** for real-time performance; released open-source.
- WaterMate – Smart Hydration System** | *ESP32, BLE, C++, Flutter, Firebase, GPT-4o-mini* | *Private* 2024 – 2026
- TÜBİTAK 2209-A funded IoT project: custom capacitive water level sensor, IMU-based drinking gesture recognition, SD card offline logging, and BLE-to-Firebase cloud sync via Flutter mobile app.
 - Integrated GPT-4o-mini chatbot for personalized hydration analysis and health recommendations.

COMPETITIONS & AWARDS

- NASA Space Apps Challenge** 2023
Global Nominee (Top 10% Worldwide) & Local 2nd Place – SHIFT Team
- Built SHIFT: a seismic data analysis and early warning platform using NASA datasets, featuring an interactive real-time earthquake monitoring dashboard and public-facing risk visualization tools.
- TEKNOFEST – Poyraz Havacilik, International UAV Competition** 2023 – 2024
Software Team Member – FINALIST
- Developed a dual-UAV autonomous inspection system for solar farms: YOLOv8/v5 defect detection (cracks, soiling), real-time image processing on Raspberry Pi, and panel health analytics reporting pipeline.
- TEKNOFEST – İLGERU UAV, Combat UAV Competition** 2023 – 2024
Software Team Member
- Implemented computer vision tasks for autonomous target detection; managed dataset annotation with Roboflow and deployed real-time YOLO inference on embedded Raspberry Pi hardware.

PUBLICATIONS

- Drone-Assisted Pallet Tracking and Counting System Using YOLOv8** IEEE, 2024
- Proposed and evaluated a UAV-based pallet tracking and counting system leveraging YOLOv8 for real-time detection in warehouse and logistics environments.
- Zero-shot Automatic Labeling in Industrial Products** 2024
- Investigated zero-shot labeling strategies for automated annotation in industrial inspection pipelines, reducing manual labeling overhead.

LANGUAGES

Turkish: Native
English: B2